Connectedness

Definition **Connectedness**

Let (X, T) be a topological space. Then X is **disconnected** if there some subsets U and V of X such that

- U and V are both nonempty and open
- $U \cup V = X$
- $U \cap V = \varnothing$.

We say X is **connected** if it is not disconnected.

Definition **Connectedness of a Subspace**

If $A \subseteq X$ is a subset in a topological space (X, T), then A is connected if it is a connected topological space given the subspace topology. That is, A is disconnected if there are some open$_X$ sets U and V such that

- $U \cup A \neq \varnothing$, and $V \cup A \neq \varnothing$
- $A \subseteq U \cup V$
- $U \cap V \cap A = \varnothing$.

Example

The space $\mathbb{R}^n$ equipped with the usual topology is connected.

However, $\mathbb{Q}$, as a subspace of $\mathbb{R}$, is disconnected. Neither is the empty set a connected subset of $\mathbb{R}$.

The integers $\mathbb{Z}$ with the cofinite topology, wherein U is open if and only if $\mathbb{Z} \setminus U$ is finite or U is empty, is connected. In fact, there are no disjoint nonempty open sets in this topology.

If (X, T) is a topological space, then $\{x\}$ is a connected subset for any $x \in X$.

Remark

In the definition of disconnectedness, the sets U and V are called **clopen**. That is, they are simultaneously closed and open.

Theorem

Suppose $f : X \to Y$ is a continuous map between two topological spaces (X, T) and (Y, U). Then the image $f(X)$ is a connected subset of Y.

Proof: We prove this by contraposition. Suppose $f(X) \subseteq Y$ is not connected. Then there are some open$_Y$ sets U and V that disconnect $f(X)$, following from the definition.

Then $f^{-1}(U)$ and $f^{-1}(V)$ are both open$_Y$ nonempty subsets of X, since f is continuous. Furthermore, $f^{-1}(U) \cup f^{-1}(V) = X$, and $f^{-1}(U) \cap f^{-1}(V) = \varnothing$. So X is not connected.

By contraposition, if X is connected, then $f(X)$ is a connected subset of Y. ■

Definition **Connected Component**

Let (X, T) be a topological space, and let $x \in X$. Then the **connected component** of x, denoted $C(x)$, is the union of all connected sets that contain x.

Example

Given $\mathbb{Q}$ endowed with the subspace topology from $\mathbb{R}$, the connected component of any $p \in \mathbb{Q}$ is simply $C(p) = \{p\}$.

If there were any rational $q \neq p$ in this set, there would be an irrational number between p and q that would admit two disjoint sets open in $\mathbb{R}$ to disconnect $C(p)$.

Theorem

Suppose (X, T) is a topological space. Then given $x \in X$, the connected component $C(x)$ is connected.

Proof: We prove this by contradiction. Suppose $C(x)$ is not connected. Then there are some open sets U and V that disconnect $C(x)$.

Without loss of generality, suppose $x \in U$. Let $y \in V \cap C(x)$. Since $y \in C(x)$, then some connected set A contains both x and y. However,

- $U \cap V \cap A \subseteq U \cap V \cap C(x) = \varnothing$
- $A \subseteq C(x) \subseteq U \cup V$
- $x \in U \cap A$, so $U \cap A \neq \varnothing$
- $y \in V \cap A$, so $V \cap A \neq \varnothing$.

Thus, A is disconnected. This is impossible, so $C(x)$ must be connected. ■

Theorem

Suppose (X, T) is a topological space. Then $C(x)$ is the *maximal* connected set containing $x \in X$. That is, there is no connected set A such that $C(x) \subsetneq A$.

Proof: We prove this by contradiction. Suppose there is a connected set A such that $x \in C(x) \subsetneq A$. Then $x \in A$, so A is in the union that defines $C(x)$. So $A \subseteq C(x)$, but this is impossible. ■

Theorem

Suppose $(X, \mathcal{T})$ is a topological space. Then given distinct points x and y in X, either $C(x) = C(y)$ or $C(x)$ and $C(y)$ are disjoint.

Proof: The notetaker implores you to accept this theorem as fact until she figures out how to draw the figures presented in lecture. ∎

Remark

In summary, connected components partition a topological space.

Theorem

Intervals in $\mathbb{R}$ with the usual topology are connected.

Proof: We prove that the interval $[0, \infty) \subseteq \mathbb{R}$ is connected. Suppose for contradiction that open sets U and V disconnect it. Without loss of generality, suppose $0 \in U$. Then V is nonempty and bounded below, so let $x = \inf V$.

Suppose $x \in U$. Then there is a nonzero radius r such that $B(x, r) \subseteq U$, so $x + r$ is a larger lower bound for V, meaning $x \neq \inf V$.

Instead, suppose $x \in V$. Then $B(x, r) \subseteq V$ for some nonzero r, meaning $x - \frac{r}{2} \in V$, so once again $x \neq \inf V$.

Either case yields a contradiction, so no such sets U and V disconnect $[0, \infty)$. Thus, it is connected. ∎

Exercise

Complete the above proof by showing that $(-\infty, 0]$ is connected.

Paths and Path-Connectedness

Definition **Path**

Let $(X, \mathcal{T})$ be a topological space. A **path** from a to b where $a, b \in X$ is a continuous map $f : [0, 1] \to X$ such that $f(0) = a$ and $f(1) = b$.

Definition **Path-Connectedness**

We call a topological space $(X, \mathcal{T})$ **path-connected** if for every pair of points $a, b \in X$, there is a path from a to b.

We write $a \sim b$ if there exists a path from a to b.

Proposition

The existence of a path between points is an equivalence relation.

Proof: Suppose (X, T) is a topological space.

- Given $x \in X$, the constant function $f : [0,1] \to X$ where $f(t) = x$ is a path from x to x. So $x \sim x$, proving reflexivity.
- Suppose $x, y \in X$. If f is a path from x to y, then we define $g : [0,1] \to X$ such that $g(t) = f(1-t)$. Then g is a path from y to x. Thus $y \sim x$, proving symmetry.
- Suppose $x, y, z \in X$. If f is a path from x to y and g a path from y to z, then we define $h : [0,1] \to X$ such that

$$h(t) = \begin{cases} f(2t) & \text{if } t \leq \frac{1}{2} \\ g(2t-1) & \text{if } t \geq \frac{1}{2} \end{cases}$$

  Then h is a path from x to z. So $x \sim z$, proving transitivity. ■

Remark

The equivalence classes of the above relation are the path-connected components of the space.

Theorem

A path-connected topological space (X, T) is connected.

Proof: Suppose for contradiction that X is path-connected but also disconnected by open sets U and V. Let $f : [0,1] \to X$ be a path from a point in U to a point in V. Then $f^{-1}(U)$ and $f^{-1}(V)$ disconnect $[0,1]$, but we know this interval is connected. So X must be connected. ■